DECLARATIVE CANVAS LAYOUT ENGINE

Sone

Build rich PDFs & images declaratively with a fluent Flexbox API

Node.js | Browser | Multi-Page PDF | Flexbox Layout

FEATURES

Column / Row / Grid
Flexbox layout containers

Text & Span
Rich inline typography

TextDefault
Inherit text styles

Photo
Images with scaleType

Path
SVG paths with fills

Table / TableRow / TableCell
Structured data tables

List / ListItem
Ordered & unordered lists

ClipGroup
SVG clip paths

Multi-Page PDF
pageHeight, header, footer

Transforms
rotate, scale, translate

Filters
blur, grayscale, invert…

QR Codes
qrcode() built-in

Sone Documentation · April 2026

01 — QUICK START

Installation & Basic Usage

Everything you need to render your first layout.

Install

```
npm install sone
# or
pnpm add sone
```

Minimal example

```
import { Column, Row, Text, sone } from 'sone';

const root = Column(
  Text("Hello, World!").size(32).weight("bold"),
  Row(
    Text("Left").flex(1),
    Text("Right"),
  ).gap(16).marginTop(12),
).padding(40).bg("white");

await sone(root).png();
// → Buffer (PNG)
```

sone() – output formats

Method	Returns
`.png()`	Promise<Buffer> – PNG image
`.jpg(quality?)`	Promise<Buffer> – JPEG (0.0 – 1.0)
`.pdf()`	Promise<Buffer> – PDF (multi-page when pageHeight set)
`.svg()`	Promise<Buffer> – SVG vector
`.webp()`	Promise<Buffer> – WebP image
`.raw()`	Promise<Buffer> – Raw pixel data
`.canvas()`	Promise<Canvas> – skia-canvas Canvas instance

SoneRenderConfig

Option	Description
`width / height`	Override canvas dimensions
`background`	Canvas background color
`pageHeight`	Enable multi-page output (px per page)
`header / footer`	Static node or (SonePageInfo) => SoneNode
`margin`	number or { top, right, bottom, left }
`lastPageHeight`	"uniform" | "content"
`cache`	Map for image caching across renders

02 — LAYOUT

Column, Row & Flexbox

All layout containers are flexbox-based. Column stacks children vertically; Row lays them out horizontally.

Column – vertical stack

```
Column(
  Text("Top"),
  Text("Middle"),
  Text("Bottom"),
).gap(8).padding(16).bg("white")
```

Top
Middle
Bottom

Children stacked top-to-bottom.

.gap(n) sets spacing between items.

.alignItems() controls cross-axis.

.justifyContent() controls main-axis.

Row – horizontal stack

```
Row(
  Text("A").flex(1),
  Text("B").flex(2),
  Text("C").flex(1),
).gap(8)
```

A flex(1) | B flex(2) | C flex(1)

.flex(n) distributes remaining space.

.shrink(0) prevents compression.

.wrap() enables line wrapping.

.alignSelf() overrides per-child.

Layout properties reference

`flex(n)`	Flex shorthand	`flexGrow / grow(n)`	Grow factor
`flexShrink / shrink(n)`	Shrink factor	`flexBasis / basis(n)`	Base size
`gap / rowGap / columnGap`	Spacing between children	`padding(…)`	CSS shorthand (1–4 values)
`margin(…)`	CSS shorthand (1–4 values)	`width / height`	Fixed or 'auto' or '%'
`size(w, h?)`	Set both at once (square if h omitted)	`minWidth / maxWidth`	Size constraints
`position()`	static | relative | absolute	`inset / top / left…`	Absolute offsets
`alignItems / alignSelf`	Cross-axis alignment	`justifyContent`	Main-axis alignment
`alignContent`	Multi-line alignment	`wrap()`	Enable flex wrapping
`overflow('hidden')`	Clip overflowing children	`display('none')`	Hide without removing from layout

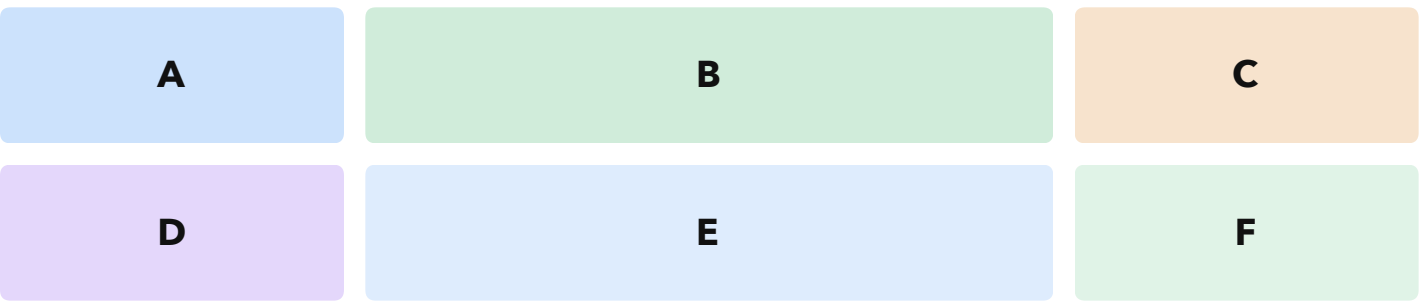
03 — GRID

CSS Grid Layout

Grid supports fixed, auto, and fractional (fr) track sizes, explicit placement, and spanning.

Basic grid

```
Grid(
  Column(Text("A")),   // auto-placed
  Column(Text("B")),
  Column(Text("C")),
)
  .columns("1fr", "2fr", "1fr")
  .columnGap(12)
  .rowGap(12)
```



Tracks: fixed(px), auto, fr

.columns(...) – column sizes

.rows(...) – row sizes

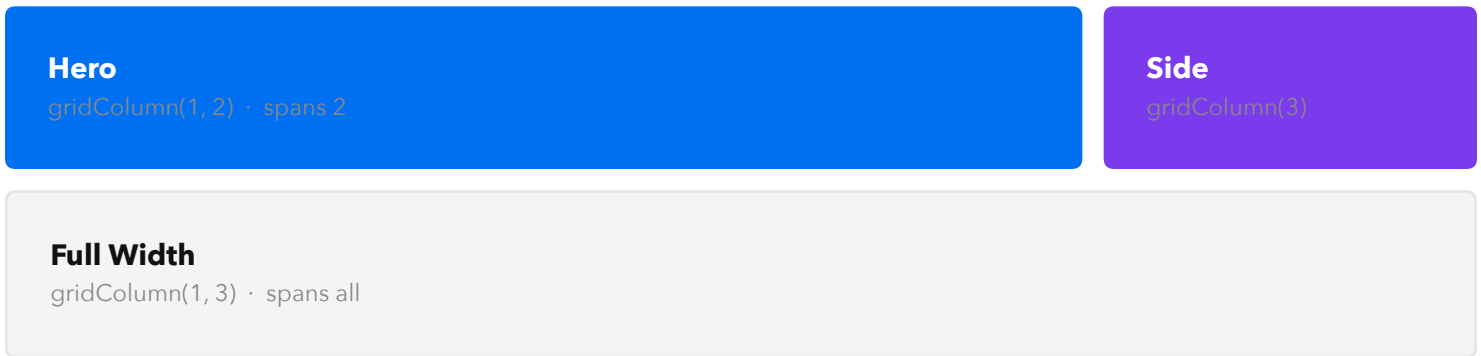
.autoRows() – implicit rows

.columnGap() / .rowGap()

Explicit placement & spanning

```
Grid(
  Column(Text("Hero")).gridColumn(1, 2).gridRow(1),  // spans 2 cols
  Column(Text("Side")).gridColumn(3).gridRow(1),
  Column(Text("Full width")).gridColumn(1, 3),      // spans all 3
```

```
).columns("1fr", "1fr", "200px")
```



.gridColumn(start, span?)

.gridRow(start, span?)

Explicit placement mixes with auto-flow.

Track size types

`150`	`"auto"`	`"2fr"`
Fixed pixels	Fit content	Fractional

04 — TEXT

Text, Span & TextDefault

Rich inline text with per-span styling, decorations, gradients, and advanced layout options.

Text – basic

```
Text("Hello world")          // plain string
Text("Hello", " world")     // multiple strings/spans
Text(Span("bold").weight("bold"), " normal")
```

Hello world

Hello world

Center

Right

Justify – this line is long enough to show justification in action here

.size(px) – font size

.color(value) – text color

.weight() – bold, 100–900

.font(...) – font stack

.style('italic') – font style

.align() – left/center/right/justify

.lineHeight(n) – line height multiplier

.lineBreak() – greedy | knuth-plass

Span – inline styling

```
Text(
  Span("Underline").underline(1.5),
  Span("Strike").lineThrough(),
  Span("Overline").overline(),
  Span("Highlight").highlight("yellow"),
  Span("Shadow").dropShadow("2px 2px 4px rgba(0,0,0,0.3)"),
```

```
  Span("Stroke").strokeColor("blue").strokeWidth(1.5),
  Span("Raised").offsetY(-6),
)
```

Underline ~~Strike~~ Overline

Highlight Shadow Stroke

Raised normal Lowered W i d e

Text layout options

`nowrap()`
Prevent line wrapping

`maxLines(n)`
Clamp visible lines

`textOverflow('ellipsis')`
Show … when clamped

`indent(px)`
First-line indent

`hangingIndent(px)`
Subsequent lines indent

`tabStops(...px)`
Tab stop positions

`tabLeader(char)`
Fill gaps e.g. "." for dot leader

`orientation(0|90|180|270)`
Layout-aware rotation

`autofit()`
Scale font to fit container

`clipImage(Photo(...))`
Image through letterforms

`wordSpacing(px)`
Extra space between words

`letterSpacing(px)`
Extra space between chars

TextDefault – inherit text styles

```
TextDefault(
  Column(
    Text("Inherits size & color"),
    Text("from TextDefault"),
  ),
).size(14).color("navy")
```

Inherits size & color

From TextDefault

Override per node

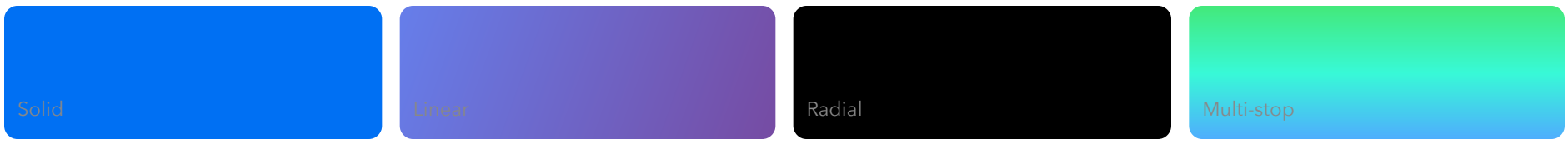
05 — STYLING

Backgrounds, Shadows, Filters & Transforms

Any layout node can have rich visual styling applied via the fluent builder API.

Background – colors, gradients, images

```
Column().bg("red")
Column().bg("linear-gradient(135deg, #667eea, #764ba2)")
Column().bg("radial-gradient(circle, yellow, orange)")
Column().bg(Photo("image.jpg").scaleType("cover"))  // image bg
```



Shadows

```
Column().shadow("0 4px 12px rgba(0,0,0,0.15)")
Column().shadow("2px 2px 0px black")        // hard
Column().shadow("0 0 20px #0070f3")         // glow
```

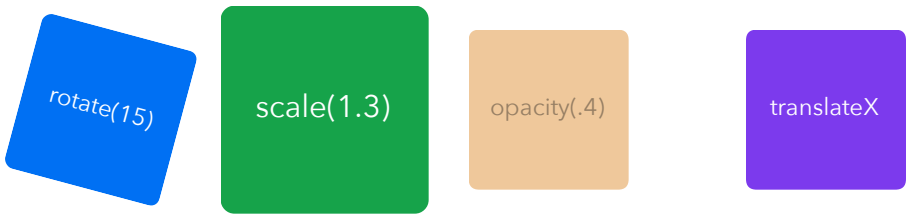
CSS Filters

```
.blur(4)          // blur(4px)
.brightness(1.4)  // brightness(1.4)
.contrast(1.5)    // contrast(1.5)
.grayscale(1)     // grayscale(1)
.invert(1)        // invert(1)
.sepia(1)         // sepia(1)
.saturate(2)      // saturate(2)
.huerotate(180)   // hue-rotate(180)
```

Transforms

```
.rotate(45)          // rotation in degrees
.scale(1.2)          // uniform scale
.scale(1.5, 0.8)     // x / y scale
```

```
.translateX(20)      // horizontal offset
.translateY(-10)     // vertical offset
.opacity(0.5)        // 0.0 – 1.0
```



06 — BORDERS & CORNERS

Per-Side Borders and Corner Styles

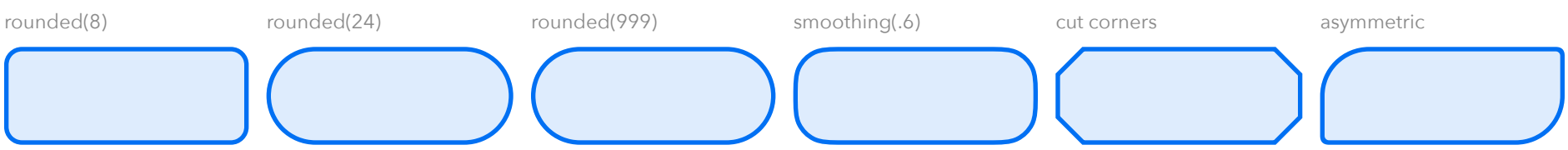
borderWidth accepts 1–4 values (CSS shorthand). Corners support radius, squircle smoothing, and cut style.

borderWidth – CSS shorthand

```
.borderWidth(2)              // all 4 sides
.borderWidth(2, 8)           // top/bottom=2, left/right=8
.borderWidth(2, 8, 4)        // top=2, left/right=8, bottom=4
.borderWidth(2, 4, 6, 8)     // top, right, bottom, left
```

Corner styles

```
.rounded(8)                  // uniform radius
.rounded(8, 0, 8, 0)         // top-left, top-right, bottom-right, bottom-left
.cornerSmoothing(0.6)        // squircle (0 – 1)
.corner("cut")               // chamfered cut corners
```



overflow('hidden') – clip children

```
Row(
  Photo("image.jpg").size(200, 100),
).rounded(16).overflow("hidden")  // clips to rounded border
```

overflow('hidden') clips

child content to the node's border-box. Works with cornerRadius as well.

07 — PHOTO, PATH & CLIPGROUP

Images, SVG Paths and Clip Groups

Render images with flexible scaling, draw SVG paths with gradient fills, and clip arbitrary shapes.

Photo – image display

```
Photo("./image.jpg").size(200, 120).scaleType("cover")
Photo(buffer).scaleType("contain").preserveAspectRatio()
Photo("icon.svg").size(32).fill("gray")  // fallback color
```

Photo props:

· src: url | Buffer

· scaleType: cover | contain | fill

· preserveAspectRatio

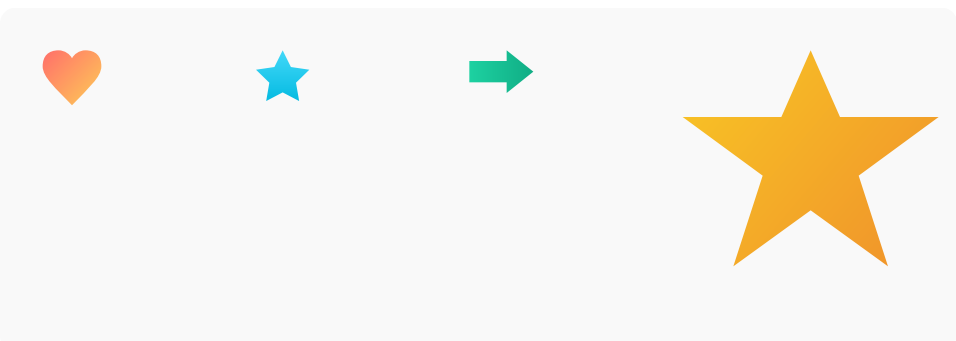
· flipHorizontal / flipVertical

· clipPath: SVG path string

· fill: placeholder color

Path – SVG path with fills

```
Path("M 0 10 L 20 0 L 40 10 L 20 20 Z")
  .fill("linear-gradient(135deg, #ff6b6b, #feca57)")
  .stroke("#333")
  .strokeWidth(2)
  .size(80, 80)
```



Path props:

· d: SVG path string

· fill: color or gradient

· stroke: color

· strokeWidth: number

· size(w, h?) – fit bounds

ClipGroup – clip children to SVG path

```
ClipGroup(
  "M 75 0 L 150 150 L 0 150 Z",   // triangle clip path
  Photo("image.jpg").size(150, 150),
```

```
  Text("Clipped!").color("white").padding(8),
).size(150, 150).overflow("hidden")
```

ClipGroup clips all children to an SVG path shape.

· Triangle, star, diamond…
· Any valid SVG path string
· Mix Photo + Text inside
· overflow('hidden') for tight clipping

08 — TABLES

Table, TableRow & TableCell

Structured data rendering with per-cell styling, column spans, and TextDefault inheritance.

Basic table

```
Table(
  TableRow(
    TableCell(Text("Name")),
    TableCell(Text("Role")),
  ).bg("#f0f0f0"),
  TableRow(
    TableCell(Text("Alice")),
    TableCell(Text("Engineer")),
  ),
).borderWidth(1).borderColor("#ccc").spacing(24, 10)
```

Name	Role	Status
Alice Chen	Engineer	Active
Bob Smith	Designer	Active
Carol Wu	Manager	Away

Table structure:

· Table(...rows)
· TableRow(...cells)
· TableCell(...nodes)

.spacing(h, v) sets cell padding.

.borderWidth / .borderColor for outer border.

Column & row spanning

```
TableCell(Text("Spans 2 cols")).colSpan(2)
TableCell(Text("Spans 2 rows")).rowSpan(2)
```

Header spans 2 cols		Col C
A1	Spans 2 rows	C1
A2		C2

.colSpan(n) – merge cols

.rowSpan(n) – merge rows

TableRow also supports:

· .bg() – row background

· .color() via TextDefault

· .weight() via TextDefault

TableCell supports:

· .alignItems() – vertical

· .bg() – cell background

09 — LISTS

List & ListItem

Ordered and unordered lists with built-in marker styles or fully custom Span markers.

Built-in styles

```
List(...items).listStyle("disc")
List(...items).listStyle("circle")
List(...items).listStyle("decimal")
```

disc

- First item
- Second item
- Third item

circle

- First item
- Second item
- Third item

decimal

1. First item
2. Second item
3. Third item

Custom Span markers

```
List(...items)
  .listStyle(Span("→").color("blue"))

// Dynamic per-item:
.listStyle((i) =>
  Span(`${i+1}.`).color(colors[i]))
```

Span marker

→ Install the package

→ Configure settings

→ Render your layout

Function marker

1. High priority task
2. Medium priority task
3. Low priority task

Nested lists

```
List(
  ListItem(
    Text("Parent item"),
    List(
      ListItem(Text("Child A")),
      ListItem(Text("Child B")),
    ).listStyle("circle").markerGap(6).gap(2).marginTop(4),
```

```
  ),
).listStyle("disc").markerGap(8).gap(8)
```

- Page Layout
  - pageHeight – enable multi-page
  - header / footer – repeating bands
  - PageBreak() – explicit breaks
- Typography
  - tabStops() – column alignment
  - orientation(90|270) – rotated text

List props:

· listStyle(style | Span | fn)

· startIndex(n) – first index for numbered

· markerGap(px) – space between marker & text

· markerOffset(px) – vertical alignment

· gap(px) – spacing between items

ListItem: any SoneNode children

including nested List nodes.

10 — MULTI-PAGE PDF

Pagination, Headers & Footers

When pageHeight is set, Sone automatically splits content across pages with optional repeating header/footer bands.

Enabling multi-page output

```
await sone(root, {
  pageHeight: 1056,           // A4/Letter page height in px
  width: 816,                // optional canvas width
  margin: { top: 16, bottom: 16 }, // space around header/footer
}).pdf();
```

Header & footer

```
// Static node — same on every page
const header = Row(
  Text("My Document").size(10),
  Text("Confidential").size(9).color("gray"),
).justifyContent("space-between").padding(8, 16)
  .borderWidth(0,0,1,0).borderColor("#e5e5e5");

// Dynamic — receives { pageNumber, totalPages }
const footer = ({ pageNumber, totalPages }: SonePageInfo) =>
  Row(
    Text("My Corp").size(9),
    Text(`${pageNumber} / ${totalPages}`).size(9),
  ).justifyContent("space-between").padding(8, 16)
    .borderWidth(1,0,0,0).borderColor("#e5e5e5");

await sone(root, { pageHeight, header, footer }).pdf();
```

pageBreak prop

Value	Effect
`"before"`	Force a page break before this element
`"after"`	Force a page break after this element
`"avoid"`	Keep element on one page – never split

```
// Explicit break node
PageBreak()  // alias for Column().height(0).pageBreak("before")

// Prop on any node
Column(...).pageBreak("avoid")
Column(...).pageBreak("before")
```

lastPageHeight

`"uniform"`
All pages have the same height. Default – required for multi-page PDF with skia-canvas.

`"content"`
Last page canvas is trimmed to actual content height. Ideal for individual page images.

11 — QR CODES & ADVANCED

QR Codes, Tab Stops, Text Orientation & Clip Image

Built-in QR code generation and advanced text layout features.

qrcode() – built-in QR generation

```
import { qrcode, Photo } from 'sone';

// Returns SVG as Buffer
const svg = qrcode("https://example.com", {
  pixelSize: 6,
  blackColor: "#000000",
  whiteColor: "#ffffff",
});

// Embed in layout as a Photo node
Photo(svg).size(120).scaleType("cover")
```

QR Code (SVG)

qrcode(data, options?) returns a Buffer containing an SVG QR code ready to embed as a Photo node.

Options:

· pixelSize – module size in px (default 4)

· blackColor – dark module color

· whiteColor – light module color

Tab stops – column alignment with \t

```
Text("Name\tScore\tRank")
  .tabStops(180, 280)       // x positions
  .tabLeader(".")           // dot leader fill
```

Name	Score	Rank
Alice Chen	9,840	1st
Bob Smith	8,120	2nd
Carol Wu	7,655	3rd

Text orientation – layout-aware rotation

```
Text("Rotated").size(16).orientation(90)  // swaps width/height in Yoga
```

clipImage – image through letterforms

```
Text("SONE")
  .size(90)
  .weight("bold")
  .clipImage(Photo("image.jpg").scaleType("cover"))
```

clipImage() fills text letterforms with an image.

The Photo is clipped to the exact shape of each glyph.

Any scaleType works: cover / contain / fill.

12 — API REFERENCE

Complete Node & Builder Reference

All exported node factory functions and their key builder methods.

Node factory functions

`Column(...children)`
Vertical flex container

`Row(...children)`
Horizontal flex container

`Grid(...children)`
CSS Grid container

`Text(...parts)`
Text block with inline spans

`Span(text)`
Inline styled text segment

`TextDefault(...children)`
Set default text styles for subtree

`Photo(src)`
Image node

`Path(d)`
SVG path node

`Table(...rows)`
Table container

`TableRow(...cells)`
Table row

`TableCell(...children)`
Table cell

`List(...items)`
List container

`ListItem(...children)`
List item

`ClipGroup(path, ...children)`
SVG clip group

`PageBreak()`
Explicit page break

`qrcode(data, opts?)`
QR code → Buffer (SVG)

`Font.load(name, src)`
Register custom font

`sone(root, config?)`
Create render context

LayoutProps – all layout nodes

`position()`	static | relative | absolute
`display()`	none | flex | contents
`bg() / background()`	Colors, gradients, images
`shadow(...css)`	CSS box shadows
`rotate(deg)`	Rotation in degrees
`translateX / translateY`	Translation
`pageBreak()`	before | after | avoid
`borderWidth(…)`	CSS shorthand 1–4 values
`rounded(…) / cornerRadius(…)`	Corner radii
`corner('cut')`	Chamfered corners
`inset / top / left / right / bottom`	Absolute position offsets
`overflow()`	visible | hidden | scroll
`opacity(0–1)`	Element opacity
`blur / brightness / contrast…`	CSS filters
`scale(x, y?)`	Scale transform
`tag(name)`	Debug identifier
`aspectRatio(n)`	Width/height ratio
`borderColor(value)`	Border color
`cornerSmoothing(0–1)`	Squircle smoothing

Sone – Declarative Canvas Layout for Node.js & the browser
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